

PAPER SUMMARY

Deep Learning with Differential Privacy



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日期
2025年10月12日

目录

CATALOGUE

01. Introduction

crowdsourced dataset
may contain sensitive
information,
causing privacy leakage.

02. Background

introduce the principle of
differential privacy and
deep learning.

03. Approach

SGD,
moments accountant
hyperparameter tuning.

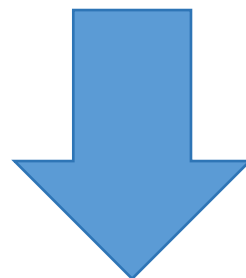
04. implementation

implementation on
MINIST
and CIFAR10

01 Introduction



crowdsourced dataset may cause data leakage



A new algorithm

means new machine
learning algorithm



a refined analysis

of privacy cost for
differential privacy

train th model
with nonconvex
objectives

Multiple objective optimization model

private deep learning

decision variable



Differential privacy restriction



Deep learning training

01

modest privacy budget(single-digit)

02

manageable cost in software complexity

03

training efficiency

04

model quality

objective function

02 Background



Differential privacy

(ϵ, δ) ϵ : privacy cost δ : probability of failure

smaller privacy cost, better privacy protection but may cause lower model precision

larger privacy cost, better model accuracy but worse privacy protection

the same query or analysis is performed on these two adjacent datasets, the probabilities of obtaining the same result should be very close



adjacent dataset

two datasets that only have one different data

Core goal: to allow statistical analysis of the dataset and mining of useful information, while ensuring that the information of individual data element in the dataset cannot be identified or inferred independently.

Deep learning

loss function: represent the penalty for mismatching the training data

θ : the average of loss on the training data



target: find the **minimum of θ**

however, the loss function of complex network is usually **nonconvex**, so it's hard to find the minimum.

this paper proposes a method — **mini-batch SGD algorithm**.



multiple basic layers

eg: simple nonlinear function



to define a function from input to output

by adjusting these parameters, to fit finite dataset

03 Approach



DP-SGD

an extension of mini-batch SGD, mixing the differential privacy and SGD algorithm.

01

Select a small batch
then compute the
original gradient

02

Gradient Clipping
clip each gradient in $\frac{\mathcal{L}}{2 \text{ norm}(???)}$

03

add noise
to obscure individual
sample

04

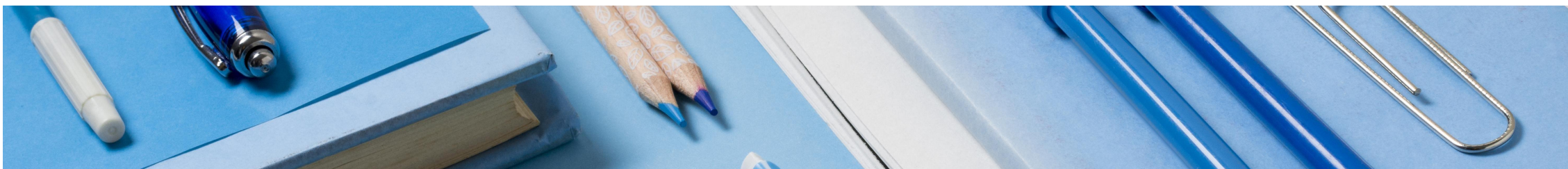
Update the network
parameters using the
"privatized gradients"
after adding noise.

To prevent the cumulative privacy loss(累计隐私预算) during the training process from exceeding the preset (ϵ , δ) constraints, the algorithm uses the **Moments Accountant** to track the privacy loss at each training step.

Compared with the traditional "Strong Composition Theorem", the **Moments Accountant** can estimate the cumulative privacy loss more **tightly**. This allows more rounds of training under the same privacy budget, **balancing** the model performance and privacy protection effect.

Hyperparameter Tuning

Satisfy with DP's limitation and get a better model



Number of Hidden Units

more hidden unit needs more noise, but do not influence the model precision.



Lot Size

the best lot size is $\sqrt{n}$
 n is the sample number



Clipping Bound, (梯度裁剪阈值)

median of the norms to control the sensitivity of gradient



Noise Level

large noise, low privacy cost allowing more epoch



Privacy Budget, ϵ

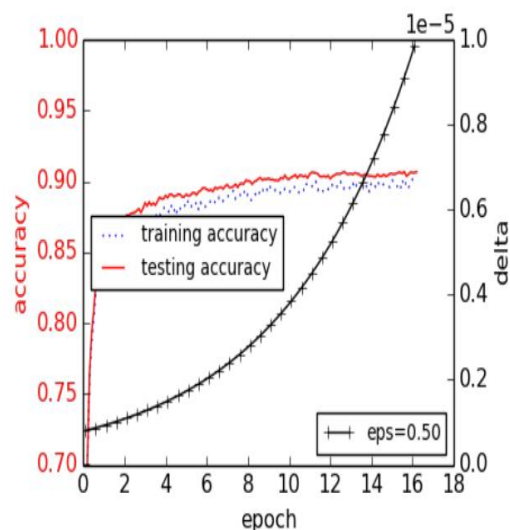
low privacy budget, stronger protection



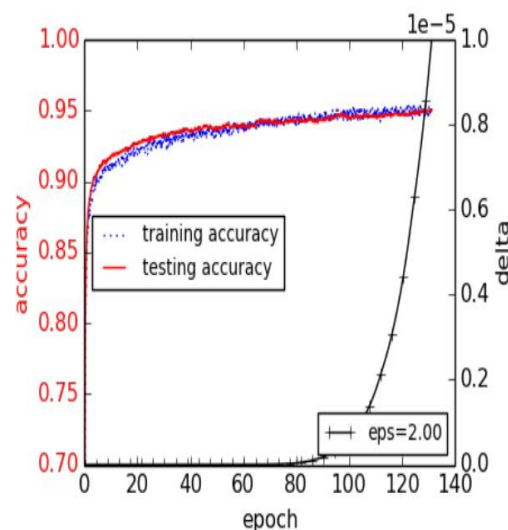
自适应调优 (? ? ?)

04 Implementation

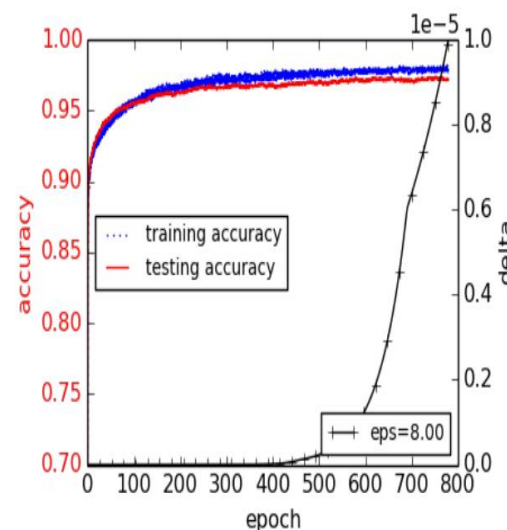




(1) Large noise



(2) Medium noise



(3) Small noise

MNIST

60,000 training examples and 10,000 testing examples, single channel, 28 × 28 size gray-level image



feedforward neural network with ReLU units



based on DP-SGD

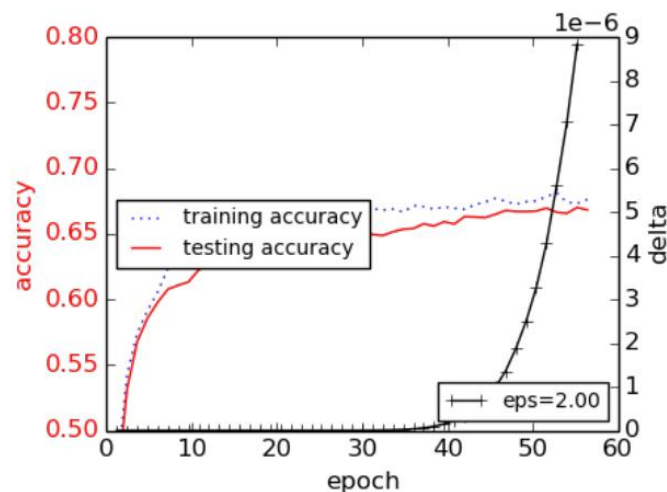
60 dimension
PCA projection

a single
1,000-unit
ReLU hidden
layer

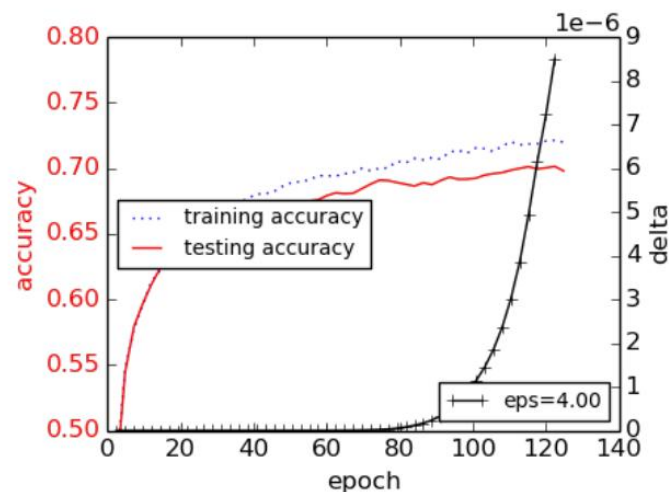
a lot size of
600
no data
augmentation

Under a "single-digit" privacy budget of only $(8, 10^{-5})$, performance close to that of non-private training is achieved

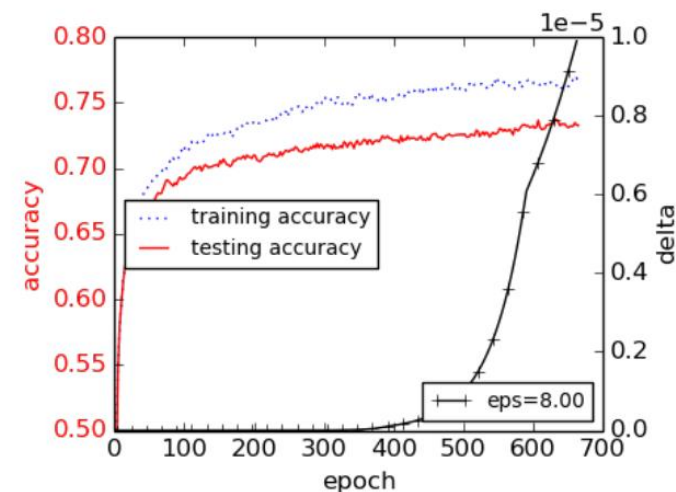
04 IMPLEMENTATION



(1) $\varepsilon = 2$



(2) $\varepsilon = 4$



(3) $\varepsilon = 8$

CIFAR10

50,000 training examples and 10,000 test examples
32 × 32 image with three channels (RGB)

two convolutional layers, two fully
connected layers(384 units)

5 × 5 convolutions, data augmentation:
Random Cropping, Flipping,
Brightness/Contrast Adjustment



Lot size:
2,000 或
4,000



gradient
clipping:
3~10



Noise
scale $\sigma=6$

Conclusion (both MNIST and CIFAR10)

Parameter sharing in convolutional layers complicates per-example gradient computation, and the **pre-training strategy** effectively mitigates this issue. (for cifar10)

A larger lot size (4,000) helps improve **stability**.



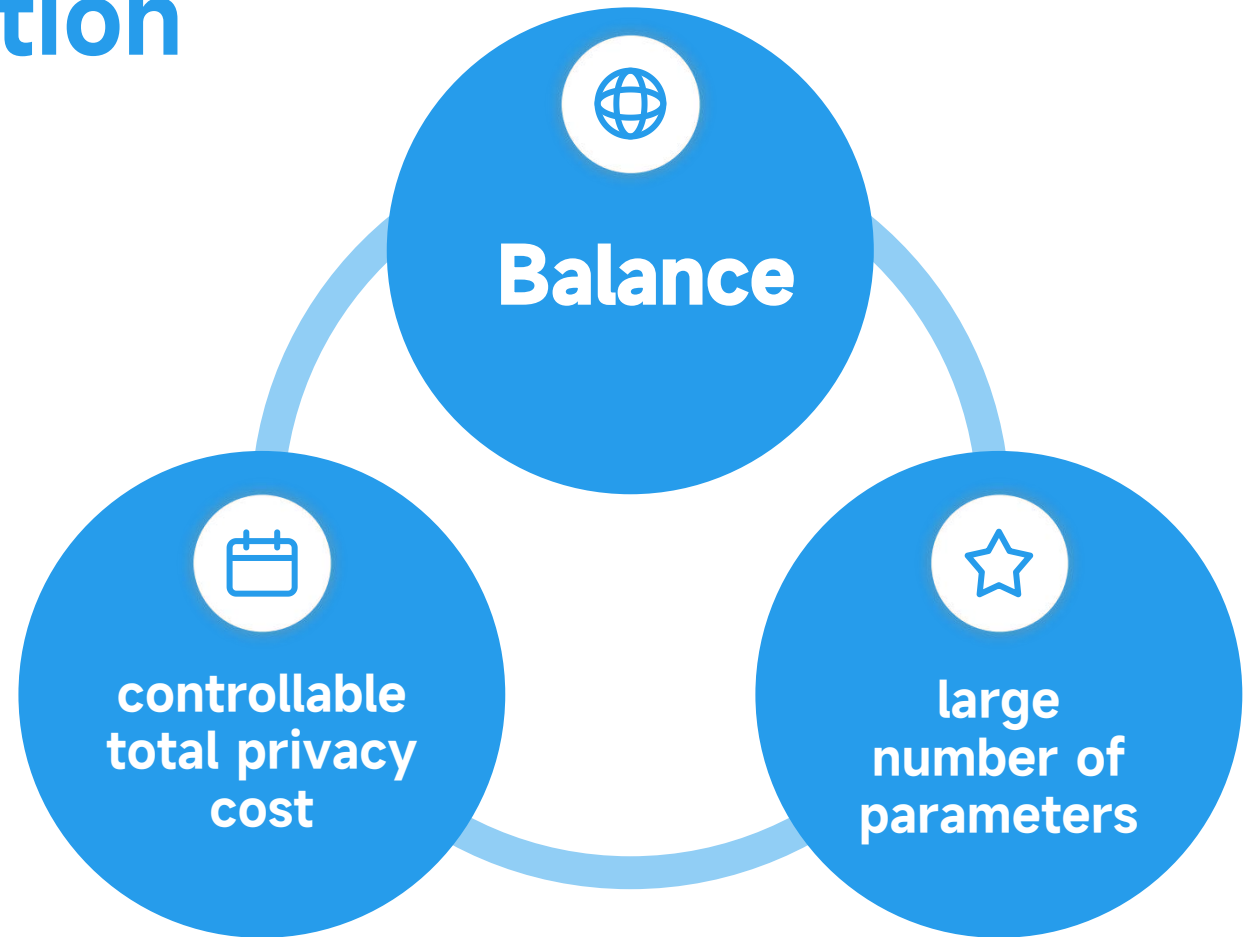
Compared with MNIST, the privacy-accuracy trade-off for CIFAR-10 is more challenging, with a more significant performance drop.

05 Innovation



Results and Contribution

The research successfully demonstrates that training **deep neural networks** with a large number of parameters is fully **feasible** under the premise of controllable total **privacy cost** (modest total privacy loss). This breaks the previous notion that deep learning (non-convex optimization) and strong privacy protection are difficult to balance.





Contribution



Moments Accountant

provide tighter privacy loss estimation



DP-SGD

The algorithm has a clear structure and can be directly integrated into mainstream frameworks such as TensorFlow, enabling privacy-preserving training with manageable software complexity and training efficiency.

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Thanks

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